

Transparency & Bank Runs: Treasury Implications

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Abstract

Is transparency a stabilizer or an accelerant in a banking crisis? Intuitively, disclosure should sustain depositor confidence, yet part of the theoretical literature shows that transparency can itself trigger a run. This paper proposes a deliberately simple model that reconciles the two views. Depositors form their confidence by combining the bank's reputation with the information it discloses, and transparency determines how much weight the disclosure receives. Transparency raises confidence when a crisis is milder than the bank's pre-crisis margin and lowers it when the crisis exceeds that margin, and its effect becomes more damaging the more severe the crisis. Whether falling confidence turns into a run depends on the bank's liquidity buffer and on the composition of its depositor base: the share of non-rational depositors and the heterogeneity of withdrawal thresholds. The model is used to discuss implications for bank and corporate treasury, and to raise open questions on how depositor heterogeneity and rationality shape the likelihood and speed of runs.

1 Introduction

In March 2023, Silicon Valley Bank lost around \$42 billion of deposits in a single day after announcing a loss on the sale of part of its securities portfolio and a plan to raise new capital that triggered the reaction in the markets. The bank was closed by regulators the following day (Board of Governors of the Federal Reserve System, 2023). A few days later, Credit Suisse, which had already experienced deposit outflows of more than CHF 110 billion in the last quarter of 2022 (Credit Suisse, 2023) was taken over by UBS in a transaction brokered by the Swiss authorities. The two episodes share a common trigger, a loss of depositor confidence, but differ in their dynamics: the first was a collapse within hours, the second an erosion over several months.

Both episodes bring a fundamental question back to the center of the debate: what role does transparency play in a banking crisis? The intuitive answer is that transparency protects a bank, since depositors who can verify that it is sound have no reason to leave. Part of the theoretical literature, however, reaches the opposite conclusion. Di Caprio and Santos-Arteaga (2012), on which this paper's intuition builds, show that imposing financial transparency on a bank whose deposit contract was designed under opacity can trigger a run with probability one, and point to the 2007 run on Northern Rock as an example. Silicon Valley Bank's case represents the perfect example because the run began precisely when the bank disclosed information that made its losses visible.

This paper argues that the two views are not in contradiction and proposes a simple model to show why. Depositors do not observe the true soundness of their bank but they form their confidence by combining its reputation with the information it discloses and transparency determines how much weight the disclosure receives. Transparency therefore acts as an amplifier of whatever it reveals. When a crisis is milder than the bank's pre-crisis margin, transparency reassures depositors and raises confidence; on the other side when the crisis exceeds that margin, transparency is the channel through which bad news reaches depositors and accelerates their exit. Moreover, the more severe the crisis, the more damaging each additional "unit of transparency" becomes.

Falling confidence, however, does not automatically produce a run. Whether it does depends on the bank's liquidity buffer and on the composition of its depositor base. The model distinguishes

between a share of non rational depositors, who withdraw regardless of the information disclosed, and a majority of rational depositors whose withdrawal thresholds differ from one another. This second dimension turns out to be of particular interest from a treasury perspective: a homogeneous depositor base appears stable until confidence crosses a common threshold and then leaves at once, while a heterogeneous base is more exposed to early withdrawals by its most nervous members but erodes more gradually. The contrast between Silicon Valley Bank and Credit Suisse fits this distinction.

The aim of the paper is not to provide a new result, but to build a small model that makes these mechanisms explicit and that can be used to reason about liquidity management and its challenges. The remainder of the paper is organised as follows. Section 2 reviews the theoretical background. Section 3 presents the model. Section 4 illustrates its main properties numerically. Section 5 discusses the implications for bank and corporate treasury, and Section 6 concludes with the limitations of the model and the open questions it raises.

2 Background

The canonical framework for bank runs is Diamond and Dybvig (1983), in which demand deposit contracts provide liquidity insurance to depositors but give rise to two equilibria: a good equilibrium in which only depositors with genuine liquidity needs withdraw early, and a run equilibrium in which all depositors withdraw because they expect others to do the same. Building on the finite-agent version of this setting developed by Peck and Shell (2003), Di Caprio and Santos-Arteaga (2012) show that when financial transparency, meaning the ability of depositors to observe the bank’s balance sheet or the withdrawal queue, is imposed after the bank has designed an optimal, incentive-compatible contract under opacity, a run occurs with probability one. Their motivating example is the 2007 run on Northern Rock, analysed in detail by O’Connor and Santos-Arteaga (2008).

Rather than asking whether transparency stabilises a bank, this paper asks under which conditions it does, focusing on the severity of the crisis and on the composition of the depositor base. To keep the analysis accessible, it abstracts from the strategic interaction between depositors (which has been demonstrated that impacts since coordination between agents can be destabilizing) and models their behavior through individual withdrawal thresholds.

3 Model

3.1 Environment

There are three dates, $t = 0, 1, 2$. A continuum of depositors of mass one each deposits one unit at a bank at $t = 0$. The bank holds a liquidity buffer $\ell \in (0, 1)$, defined as the share of deposits it can pay out at $t = 1$ from cash, highly liquid assets and asset sales, and invests the remainder in illiquid assets that pay off at $t = 2$. Depositors may withdraw at $t = 1$ or wait until $t = 2$. Denoting by w the share of depositors withdrawing at $t = 1$, the bank fails if

$$w > \ell. \tag{1}$$

3.2 Fundamentals, crisis and disclosure

The bank’s true soundness is

$$\theta = d - s, \tag{2}$$

where d measures how sound the bank was before the crisis relative to its reputation, and $s \geq 0$ is the severity of a crisis shock, such as a rise in interest rates that erodes the value of the bank’s

securities portfolio. Depositors do not observe θ , in fact their prior opinion of θ , which reflects the bank's reputation, is normalised to $\theta \sim \mathcal{N}(0, 1)$; in other words the crisis is not yet reflected in public perception. The bank then discloses a signal

$$y = \theta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1/\tau), \quad (3)$$

where y is the signal as the sum of the true soundness of the bank and an error component σ with mean 0 (the bank doesn't systematically lie and variance as $1/\tau$; where $\tau \geq 0$ is the bank's *transparency*, measured as the precision of its disclosure relative to the precision of its reputation. Under full opacity ($\tau = 0$) the disclosure is non-informative (meaning that that variance would spread infinitely); under full transparency ($\tau \rightarrow \infty$) it reveals θ exactly (the variance would collapse around zero).

By Bayes' rule, depositors' *confidence*, defined as their posterior estimate of the bank's soundness, is

$$F = \mathbb{E}[\theta | y] = \omega(\tau) y, \quad \omega(\tau) = \frac{\tau}{1 + \tau}. \quad (4)$$

Confidence is a weighted average of reputation (normalised to zero) and disclosed information, and transparency determines the weight ω that depositors place on the disclosure.

Since the disclosure is unbiased ($\mathbb{E}[y] = \theta$), expected confidence is $\mathbb{E}[F] = \omega(\tau) (d - s)$, and

$$\frac{\partial \mathbb{E}[F]}{\partial \tau} = \frac{d - s}{(1 + \tau)^2}, \quad \frac{\partial^2 \mathbb{E}[F]}{\partial \tau \partial s} = -\frac{1}{(1 + \tau)^2} < 0. \quad (5)$$

Transparency raises confidence when the crisis is milder than the bank's pre-crisis margin ($s < d$) and lowers it when the crisis exceeds that margin ($s > d$). Moreover, the more severe the crisis, the more damaging each additional unit of transparency becomes. Transparency is the channel through which the crisis reaches depositors: under opacity the crisis remains hidden, under full transparency it is fully revealed.

3.3 Depositors

A share $\lambda \in [0, \ell]$ of depositors is *non-rational*: they withdraw at $t = 1$ regardless of the information disclosed, out of panic or imitation. The remaining $1 - \lambda$ are rational but heterogeneous: depositor i withdraws if confidence falls below a personal threshold $c_i \sim \mathcal{N}(\bar{c}, \sigma^2)$. The mean $\bar{c}$ captures how demanding depositors are on average, while σ shows the heterogeneity of the depositors. A low σ describes a concentrated, homogeneous clientele. The share of depositors withdrawing is therefore

$$w(F) = \lambda + (1 - \lambda) \Phi\left(\frac{\bar{c} - F}{\sigma}\right), \quad (6)$$

where Φ denotes the standard normal distribution function.

3.4 Run condition

Combining (1) and (6), the critical level of confidence F^* below which a run occurs is implicitly defined by

$$\lambda + (1 - \lambda) \Phi\left(\frac{\bar{c} - F^*}{\sigma}\right) = \ell. \quad (7)$$

The left-hand side is the share of depositors withdrawing when confidence equals F^* , the right-hand side is the share the bank can pay. The lower F^* , the more resilient the bank, since confidence must fall further before a run is triggered. Equation (7) makes the main drivers of fragility transparent:

- A larger ℓ lowers F^* : the bank can absorb more withdrawals so confidence must deteriorate even more before it fails. So having the correct liquidity buffer is essential.

- A larger λ raises F^* meaning that part of the buffer is consumed by depositors who leave regardless (non-rational ones), leaving less room for rational ones.
- A higher $\bar{c}$ raises F^* : depositors withdraw at higher levels of confidence because they are more demanding.
- The effect of σ depends on the buffer. When the bank can afford to lose less than half of its rational depositors, which is the realistic case, greater heterogeneity raises F^* : the most nervous tail of the distribution alone is enough to consume the buffer. A homogeneous depositor base is instead stable until confidence reaches $\bar{c}$, at which point almost all depositors withdraw at once. Heterogeneity therefore trades a higher probability of a run for a more gradual one.

Under full transparency ($\tau \rightarrow \infty$), confidence coincides with the bank's true soundness, $F = \theta$, and a bank with $\theta < F^*$ faces a run with probability one as demonstrated by Di Caprio and Santos-Arteaga (2012) in this simplified setting.

Table 1: Notation

Symbol	Meaning	Domain
ℓ	Liquidity buffer	$(0, 1)$
w	Share of depositors withdrawing at $t = 1$	$[0, 1]$
d	Pre-crisis soundness relative to reputation	$\mathbb{R}$
s	Crisis severity	$[0, \infty)$
θ	True soundness of the bank	$\mathbb{R}$
τ	Transparency (precision of disclosure)	$[0, \infty)$
ω	Weight placed on the disclosure	$[0, 1)$
F	Depositor confidence	$\mathbb{R}$
λ	Share of non-rational depositors	$[0, \ell)$
$\bar{c}$	Average withdrawal threshold	$\mathbb{R}$
σ	Heterogeneity of the depositor base	$(0, \infty)$
F^*	Critical confidence level	$\mathbb{R}$

4 Results

This section illustrates the main properties of the model numerically. The parameter values are chosen for illustration purposes and are not calibrated to any specific bank. For this specific illustration, the pre-crisis margin is $d = 0.5$, the liquidity buffer is $\ell = 0.30$, the share of non-rational depositors is $\lambda = 0.05$ and withdrawal thresholds have mean $\bar{c} = -0.5$ and dispersion $\sigma = 0.5$. These values imply a critical confidence level $F^* \approx -0.18$. The full model can be found in the link in the footnote so that anyone can calibrate the parameters as one prefers.¹

Figure 1 shows the expected confidence against transparency for a mild and a severe crisis. In the mild crisis, which is smaller than the bank's pre-crisis margin, confidence increases with transparency: the more depositors rely on the bank's disclosure, the more they are reassured. In the severe crisis the pattern is reversed. Confidence falls as transparency rises and crosses the threshold at $\tau^* \approx 0.58$; beyond this level, additional disclosure pushes the bank into the run area. Both curves flatten as transparency grows, because once depositors rely mostly on the disclosure, additional precision changes their beliefs little. The figure in particular plots the central message

¹The code used to generate the figures together with an interactive model in which every parameter of the model can be varied, is available at <https://github.com/mafantin/bank-run-transparency> just by clicking the link at the beginning of the README file.

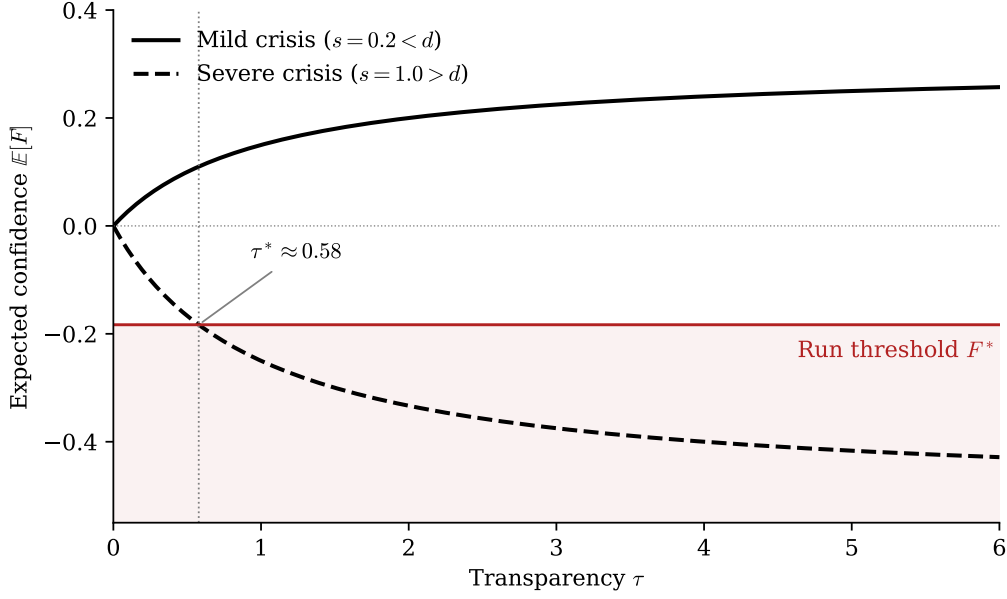


Figure 1: Expected confidence as a function of transparency. Mild crisis ($s = 0.2$) Severe crisis ($s = 1$). The shaded area is the run region, below the critical confidence level F^* .

of equation (5): the level of transparency that sustains confidence cannot be chosen independently from the state of the crisis.

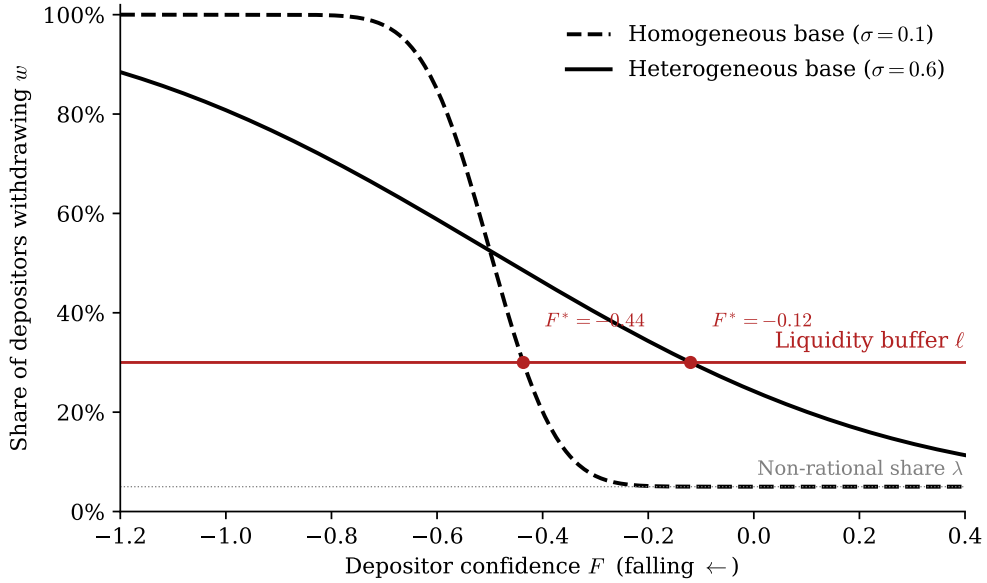


Figure 2: Share of depositors withdrawing as a function of confidence, homogeneous ($\sigma = 0.1$) heterogeneous ($\sigma = 0.6$) depositor base. Red dots mark the critical confidence level F^* at which withdrawals consumes the liquidity buffer ℓ .

Figure 2 plots the share of depositors withdrawing as a function of confidence and is read from right to left. With a homogeneous depositor base, withdrawals remain at the level of non-rational depositors until confidence approaches the common threshold and then rise almost vertically. The buffer is consumed only when confidence falls to $F^* \approx -0.44$, but when this happens nearly all depositors leave almost instantly. With a heterogeneous base, withdrawals start at much higher levels of confidence from the most nervous depositors and consume the buffer earlier, at $F^* \approx -0.12$,

but they build up gradually without a cliff. A homogeneous base therefore makes a run less likely but violent while on the other side, a heterogeneous base makes it more likely but slower. The first pattern resembles the collapse of Silicon Valley Bank within a single day, the second the erosion of Credit Suisse’s deposits over several months.

Taken together the two figures separate the two stages of a run. Transparency and the severity of the crisis determine IF confidence falls while the liquidity buffer and the composition of the depositor base determine WHEN and HOW FAST falling confidence turns into a run. Both stages are, to a large extent, under the control of a bank’s treasury as discussed in the next section.

5 Implications for Treasury

Each component of this model corresponds to a variable that a bank’s treasury monitors or manages. This section discusses three implications.

The first implication refers to the liquidity buffers that should reflect the composition of the depositor base, not only its size. In the model, the strenght of the bank is captured by the critical confidence level F^* , which depends not only on the size of the liquidity buffer ℓ but also on how depositors are distributed. Regulatory liquidity requirements already differentiate across depositor categories: under the Liquidity Coverage Ratio, assumed run-off rates range from a few percent for stable, insured retail deposits to 40% or more for non-operational wholesale deposits (Basel Committee on Banking Supervision, 2013). These rates, however, capture how likely each category is to leave, not how jointly its members are likely to leave. The model suggests that this second dimension is relevant. A homogeneous depositor base, such as one concentrated in a single sector or network of clients, appears stable in normal times and then withdraws almost entirely once confidence crosses a common threshold. For treasury management, indicators such as depositor concentration, the share of uninsured deposits and the sectoral composition of the client base should therefore be seen as extremely relevant measures of liquidity risk. The heterogeneity result also clarifies what diversification of the funding base can and cannot achieve: in the realistic case of a small buffer, a more heterogeneous base does not make a run less likely, but it makes it slower. Slower outflows buy time to search and activate contingency funding, such as repo lines or central bank facilities, an that the static model does not capture but that is central to treasury.

The second implication shows that short term deposits could consume all the buffer before confidence is even tested. Non-rational depositors in the model withdraw regardless of the bank’s condition and every unit of them reduces the buffer available to absorb rational withdrawals. In reality, deposits attracted by promotional rates, held through digital channels play a similar role. A treasury that grows its deposit base through such channels increases λ and with it, it also grows the level of confidence at which a run is triggered, even if the average quality of its depositors appears unchanged.

The third and last implication discussed says that in a crisis, liquidity should come first with respect to disclosure. Equation (5) implies that when a crisis exceeds the bank’s pre-crisis margin, greater transparency lowers confidence. This is not an argument for concealment: disclosure obligations are largely beyond a bank’s discretion and opacity would erode the reputation on which confidence rests. The implication concerns sequencing. Because disclosure transmits bad news to depositors, the liquidity buffer should be adapted to the situation before that news is released, by securing funding lines and collateral with the central bank, rather than after.

6 Limitations and Open Questions

The simplicity of this model comes at the cost of several assumptions. Withdrawal thresholds are exogenous: depositors do not react to what they expect others to do, so the model abstracts from the aspect at the heart of Diamond and Dybvig (1983). Reputation does not yet incorporate the crisis, meaning that all information about it flows through the bank's disclosure, whereas in reality markets and the press also transmit news. The model is static: it determines whether a run occurs, but not how fast it unfolds, only the graduality is shown. The liquidity buffer is a single number that already includes asset sales and there is no deposit insurance. Finally, all parameter values are illustrative. These limitations point directly to the questions the model leaves open.

Does the content of disclosure matter, and not only its precision? Transparency in the model is a single parameter: the bank discloses an unbiased signal about its overall soundness and τ measures only how precise that signal is. In practice, banks disclose a large amount of information under regulatory requirements, but retain discretion over what goes beyond them. Some information, such as internal assessments or planned transactions is not disclosed at all. When such information later reaches the public through other channels, it can move confidence and trigger a run that the bank's own disclosure had not anticipated.

How much does heterogeneity matter? In the model, the dispersion of withdrawal thresholds changes both the confidence level at which a run starts and the way it develops, independently of how confident depositors are on average. Whether this effect is large in practice is an empirical question. Comparing banks with similar liquidity ratios but different degrees of deposit heterogeneity, and asking whether heterogeneity predicts the speed of outflows under stress, would be an interesting test.

Is a more rational clientele a safer one? The model treats non-rational depositors as the source of fragility. The case of Silicon Valley Bank suggests a less comfortable possibility. Its depositors were sophisticated firms, well informed and closely connected through the same investors and advisers. Each acted rationally, but shared information aligned their thresholds, producing precisely the homogeneous base that the model identifies as more likely to total runs. If the clientele is rational and homogeneous due to connectedness it may be the case that is not less fragile than a heterogeneous clientele (mixed rational and non-rational) and this could mean that the degree of connectedness among depositors may deserve a place in liquidity risk assessment.

How does speed change the picture? Mobile banking and social media have compressed the time in which a loss of confidence becomes a run from days to hours. In a dynamic version of the model, the liquidity buffer would matter not only for its size but for the time it buys.

How does deposit insurance reshape the depositor base? Insurance removes the incentive to run for insured depositors, shrinking the population that can withdraw. The depositors who remain exposed are typically the largest and most concentrated ones. Insurance may therefore reduce the risk of panic while making the remaining runnable deposits more homogeneous.

Code availability

All figures can be reproduced with the Python code available at <https://github.com/mafantin/bank-run-transparency>. The repository also contains an interactive model in which every parameter of the model can be varied, allowing the reader to explore the behaviour of the model beyond the cases discussed in the text.

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